note0403

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Often, the number of genetic marker covariates (m) included in genomic models exceeds the number of records (n). When the genomic model is parameterised in terms of genetic marker effects that are the focus of interest, given an $m \gg n$ setup, some form of regularisation is needed in order to obtain a solution for the m genetic marker effects using the n records available.

Consider a standard genomic model where y, an $n \times 1$ column vector of data has zero mean. The hierarchical structure of the model is as follows:

$$y \mid \beta, \sigma^2 \sim N\big(X\beta,\, I\sigma^2\big)\,, \tag{1a}$$

$$\beta \mid \sigma^2_\beta \sim N\big(0,\, I\sigma^2_\beta\big)\,. \tag{1b}$$

Regularisation is introduced by assigning a distribution to the vector β. The column vector of marker effects β, the focus of inference, is of dimension $m \times 1$ and in a first stage, it is assumed that the variance components σ^2 and σ^2_β are known so that β is the only unknown parameter. Matrix X is often rank deficient: $\text{rank}(X) = r$, $r \leq n$.

This note is in four parts: the first part describes how the data vector y contributes to the inference about the vector of marker effects β in this particular setup where the likelihood depends on β only through $X\beta$. The basic idea is to express the vector of marker effects as the sum of two orthogonal linear combinations of the elements of β: one that is identifiable and the other that is unidentifiable by the data. The second part examines the process of Bayesian learning under two scenarios: one where variance components are known and the other where they are jointly inferred with the genetic marker effects β. The third and fourth parts discuss how identifiability of the vector of marker effects plays out in the context of the classical ridge regression and the lasso, respectively.

Preliminaries

1. Use will be made of the Moore-Penrose inverse (Searle, 1982). Given a matrix X, the Moore-Penrose inverse X^+ is a *unique* matrix that satisfies the following criteria:

   - $XX^+X = X$
   - $X^+XX^+ = X^+$

- XX^+ is symmetric
- X^+X is symmetric

Given X, there are many matrices G that satisfy the first of these conditions. This is in contrast with the Moore-Penrose inverse that is unique.

2. Singular value decomposition. In general, if X is an $(n \times m)$ matrix of rank r, then it can be written as
$$X = UDV', \tag{2}$$
where

   - U is an $n \times r$ matrix whose columns are orthonormal left singular vectors of X;
   - $D = \text{diag}(d_1, \ldots, d_r)$ is an $r \times r$ diagonal matrix whose diagonal elements satisfy $d_1 \geq d_2 \geq \cdots \geq d_r > 0$ and are the positive singular values of X;
   - V is an $m \times r$ matrix whose columns are orthonormal right singular vectors of X.

   The columns of U and V are orthonormal, that is, $U'U = I_r$, $V'V = I_r$, and unless $r = n$, $UU' \neq I_n$; rather, UU' is the orthogonal projector onto the column space of X. Similarly, unless $r = m$, $VV' \neq I_m$; instead, VV' is the orthogonal projector onto the row space of X.

3. Define the $m \times m$ projection matrix P_R as
$$\begin{aligned} P_R &= X'(XX')^+X \\ &= X^+X \end{aligned} \tag{3}$$
where $(XX')^+$ is the Moore-Penrose inverse of XX' and X^+ is the Moore-Penrose inverse of X. Matrix P_R is the orthogonal projector of vectors onto the row space of matrix X. This means that $\beta_R = P_R\beta$ lies in the row space of X, whereas $\beta_N = (I - P_R)\beta$ lies in the null space of X. Then,
$$\begin{aligned} X\beta_N &= X(I - P_R)\beta \\ &= X\beta - XX^+X\beta \\ &= 0 \end{aligned} \tag{4}$$
because $XX^+X = X$, due to the first Penrose identity.

   The proof of equality (3) is as follows. From the partitioning (2),
$$XX' = UDV'VDU' = UD^2U' \Rightarrow (XX')^+ = UD^{-2}U',$$
therefore
$$X'(XX')^+X = VDU'(UD^{-2}U')UDV' = VV'.$$

Inverting (2)

$$X^+ = VD^{-1}U' \Rightarrow X^+X = VD^{-1}U'UDV' = VV',$$

establishing the equality. In this expression, $D^{-1} = \text{diag}\left(\frac{1}{d_1}, \ldots, \frac{1}{d_r}\right)$, where $r = \text{rank}(X)$.

When the projection matrix is computed with the Moore-Penrose inverse, $P_R = P_R^2$ and $P_R' = P_R$; that is, P_R is idempotent and symmetric.

Partitioning the vector of genetic marker effects

The vector β can be expressed as the sum of two orthogonal components

$$\begin{aligned} \beta &= P_R\beta + (I - P_R)\beta \\ &= \beta_R + \beta_N, \end{aligned} \tag{5}$$

where $\beta_R'\beta_N = 0$ (NOTE: $\beta_R'\beta_N = \beta' P_R'(I - P_R)\beta = \beta' 0 \beta = 0$, using $P_R = P_R'$, $P_R^2 = P_R$). Then,

$$\begin{aligned} X\beta &= X\beta_R + X\beta_N \\ &= X\beta_R \end{aligned}$$

due to (4). Therefore the model $y = X\beta + e$ can be written as $y = X\beta_R + e$: data y contain information about the row space component, β_R, but not about the null space component, β_N. When the likelihood depends on β only through $X\beta$, the fraction of β that lies in the null space of X, $(I - P_R)\beta$, is not identifiable from y.

When the rank of X is $r = m$, $VV' = I_m$. In this case $P_R = I_m$, $(I_m - P_R) = 0$, and $\beta_N = 0$, indicating that the null space has disappeared; the entire vector β is identifiable from the data. Then

$$P_R\beta = \beta.$$

Parametrising the genomic model in terms of genomic values

Assume that instead of parametrising the genomic model in terms of marker effects β, the model takes the form:

$$y \mid g, \sigma^2 \sim N\left(g,\, I\sigma^2\right), \tag{6a}$$

$$g \mid \sigma_G^2 \sim N\left(0,\, G\sigma_G^2\right) \tag{6b}$$

where $G = \frac{XX'}{m}$ and $\sigma_G^2 = m\sigma_\beta^2$. The covariance matrix $G\sigma_G^2$ is positive semidefinite (possibly singular), so g has a singular multivariate normal distribution whose support is the column space of X, whenever $\text{rank}(X) < n$, $m > n$. Despite the rank deficiency of X, under this parametrisation, every admissible value of the parameter g is identifiable, as shown below.

Define $P_C = XX^+$, the symmetric and idempotent $n \times n$ matrix that projects vectors onto the column space of X. Notice that this is in contrast with the $m \times m$ matrix P_R, a projector onto the row space of X. Then, with $g = X\beta$,

$$\begin{aligned} P_C\, g &= XX^+X\beta \\ &= X\beta = g \end{aligned}$$

because $XX^+X = X$, due to the first Penrose identity. Every admissible parameter g lies in the column space of X; thus the parameter space coincides with the identifiable parameter space and the projector P_C is the identity operator. There is no unidentifiable component of g, in contrast with the partitioning in (5): for β, the parameter space is $\mathbb{R}^m$ but only the projection onto the row space of X is identifiable.

The second part of this note examines how identifiability of β plays out when matrix X of the genomic model (1) is also rank deficient but variance components σ^2_β and σ^2 are unknown and updated by the data.

Bayesian learning: updating genetic marker effects

The question addressed is whether, given a hierarchical Bayesian model and a body of data, Bayesian inference leads to learning about all components of vector β. This is investigated by comparing the prior and posterior distributions of the marker effects. If the posterior distribution of a component is identical to its prior distribution, the data provide no information about that component beyond that already contained in the prior, and no Bayesian learning has occurred for that component.

Variance components known

Consider again the standard genomic model that is described by the following hierarchy:

$$y \mid \beta, \sigma^2 \sim N\big(X\beta,\, I\sigma^2\big)\,, \tag{7a}$$

$$\beta \mid \sigma^2_\beta \sim N\big(0,\, I\sigma^2_\beta\big)\,. \tag{7b}$$

The vector y contains the n phenotypic observations, X is the observed $(n \times m)$ marker matrix with $m > n$ and rank $r < \min(n, m)$, and β is the unobserved $(m \times 1)$ vector of marker effects. The variance components σ^2 and σ^2_β are assumed known.

As indicated in the first part of this note, the vector of genetic marker effects can be partitioned into a component that is identifiable from the data, β_R, and an orthogonal component that is not identifiable from the data, β_N. The likelihood for the genomic model can be written as $p(y|\beta_R, \beta_N, \sigma^2) = p(y|\beta_R, \sigma^2)$ because $X\beta_N = 0$, and therefore the posterior distribution of β takes the following form:

$$p\left(\beta_R, \beta_N | y, \sigma^2, \sigma^2_\beta\right) \propto p(y|\beta_R, \sigma^2)p\left(\beta|\sigma^2_\beta\right).$$

As shown in the NOTE below, given the prior in (7b), the prior density of vector β factorises as $p\left(\beta|\sigma_\beta^2\right) = p\left(\beta_R|\sigma_\beta^2\right) p\left(\beta_N|\sigma_\beta^2\right)$; therefore, since the likelihood does not include β_N, the posterior distribution of β_N, after integrating over β_R, is given by

$$p\left(\beta_N|y, \sigma^2, \sigma_\beta^2\right) \propto p\left(\beta_N|\sigma_\beta^2\right),$$

indicating absence of Bayesian learning for the non-identifiable component β_N.

NOTE

Under normality, with $\beta|\sigma_\beta^2 \sim N(0, I_m\sigma_\beta^2)$ and $\beta_R = P_R\beta$, $\beta_N = (I_m - P_R)\beta$,

$$\begin{aligned} Cov(\beta_R, \beta_N'|\sigma_\beta^2) &= Cov(P_R\beta, \beta'(I_m - P_R)|\sigma_\beta^2) \\ &= P_R(I_m - P_R)\sigma_\beta^2 = 0 \end{aligned}$$

since P_R is symmetric and idempotent. Therefore $p\left(\beta|\sigma_\beta^2\right) = p\left(\beta_R|\sigma_\beta^2\right) p\left(\beta_N|\sigma_\beta^2\right)$.

Variance components unknown

Assume that σ_β^2 is unknown and updated from the data, and for simplicity, assume σ^2 is known. The posterior distribution of β_N takes the following form:

$$p\left(\beta_N|y, \sigma^2\right) = \int p\left(\beta_N|y, \sigma_\beta^2, \sigma^2\right) p\left(\sigma_\beta^2|y\right) d\sigma_\beta^2.$$

Since $p\left(\beta_N|y, \sigma_\beta^2, \sigma^2\right) = p\left(\beta_N|\sigma_\beta^2\right)$, the posterior distribution becomes

$$p\left(\beta_N|y, \sigma^2\right) = \int p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2|y, \sigma^2\right) d\sigma_\beta^2.$$

This expression is different from

$$\int p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2\right) d\sigma_\beta^2$$

unless $p\left(\sigma_\beta^2|y, \sigma^2\right) = p\left(\sigma_\beta^2\right)$ which is usually not the case since the data updates σ_β^2. However, the mean of the posterior distribution of β_N remains unchanged relative to the prior mean. Indeed,

$$\int\int \beta_N\, p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2|y, \sigma^2\right) d\beta_N d\sigma_\beta^2 =$$
$$\int \left[\int \beta_N\, p\left(\beta_N|\sigma_\beta^2\right) d\beta_N\right] p\left(\sigma_\beta^2|y, \sigma^2\right) d\sigma_\beta^2 = 0$$

since the integral in square brackets corresponds to the prior mean.

Conclusion: when β and σ_β^2 are inferred from a joint Bayesian analysis, the nonidentifiable component β_N of β undergoes indirect Bayesian learning through the posterior updating of the variance component σ^2_β. The likelihood remains completely flat in the β_N direction, so the identifiability structure of the model is unchanged: β_N is still not directly identifiable from the data. These results and those presented in the first part of this note, highlight the distinction between inference of genomic values $g = X\beta$ and inference of marker effects β. Genomic values are identifiable whereas only the β_R component of β is learned directly from the data. Consequently in genomic models where $m \gg n$, inference of individual marker effects is inherently more challenging than inference of genomic values.

How Does the Penalty Term in Ridge Regression Alter the Identifiability of b?

The first part of this note discussed identifiability of the parameter β of the linear regression model $y = X\beta + e$ when matrix X is rank deficient. It was indicated that the vector of genetic marker effects β can be partitioned into two orthogonal linear combinations: β_R that is identifiable by the data y, and β_N that is not (see page 3). The second part of this note shows how a Bayesian implementation deals with the rank deficiency of matrix X in order to infer β. When variance components are known and the only unknown parameter is β, given a Gaussian prior distribution for β of the form $N(0, I\sigma^2_\beta)$, the Bayesian analysis updates β_R via the data, but the information to infer the unidentifiable component β_N originates solely from the prior distribution. The posterior mode of β_N (and given the Gaussian assumption, the posterior mean) is equal to zero. This third part illustrates how this process unfolds in the context of ridge regression.

Referring to page 3, partition β into its two orthogonal components:

$$\beta = \beta_R + \beta_N. \tag{8}$$

The ridge coefficients are obtained as

$$\hat{\beta} \underset{\arg\min\, \beta}{=} (y - X\beta)' (y - X\beta) + \lambda\beta'\beta,$$

where here, the shrinkage parameter $\lambda > 0$. Using (8), the orthogonality of β_R and β_N, and the equality $X\beta = X\beta_R$, the cost function of the ridge regression is

$$J(\beta_R, \beta_N|\lambda) = (y - X\beta_R)'(y - X\beta_R) + \lambda\beta'_R\beta_R + \lambda\beta'_N\beta_N.$$

For any fixed value of β_R, the first two terms are constant with respect to β_N. Therefore

$$\hat{\beta}_N \underset{\arg\min\, \beta_N}{=} \lambda\, \beta'_N\beta_N.$$

Since $\lambda > 0$, $\lambda\, \beta'_N\beta_N \geq 0$, with equality only if $\beta_N = 0$. Thus,

$$\hat{\beta}_N = 0. \tag{9}$$

The remaining component $\hat{b}_R$ is obtained by minimising

$$(y - X\beta_R)'(y - X\beta_R) + \lambda\, \beta'_R\beta_R.$$

Conclusion: Ridge regression only estimates the component of β that is identifiable by the data: the data determine only β_R. The unidentifiable component β_N is set to zero because this minimises the cost function: the penalty determines β_N. The solution for β_N agrees with the Bayesian approach under the Gaussian prior for β: the posterior distribution of β_N is equal to its prior distribution, whose mean (and mode) is zero.

How Does the Penalty Term in Lasso Alter the Identifiability of b?

This section summarises the concept of identifiability of genetic marker effects b, when the matrix of observed marker effects X is rank deficient and lasso is used as the operational model. As before, partition vector b into its orthogonal components

$$b = b_R + b_N. \tag{10}$$

Then, since b_N belongs to the null space of X, $Xb = Xb_R$, and the loss function associated with lasso can be written as

$$J(b_R, b_N|\lambda) = (y - Xb_R)'(y - Xb_R) + \lambda \left\| b_R + b_N \right\|_1 \tag{11}$$

where the last term in the right hand side is the lasso's ℓ_1-norm penalty. In the case of ridge regression, $\|b_R + b_N\|_2^2 = \|b_R\|_2^2 + \|b_N\|_2^2$, which upon minimisation leads immediately to $\hat{b}_N = 0$, regardless of the value of b_R. However, this factorisation does not hold for the last term in (11) despite the orthogonality of b_R and b_N. Therefore, for a fixed value of b_R, minimisation of the loss function (11) with respect to b_N, where b_N belongs to the null space of X, does not necessarily imply $\hat{b}_N = 0$.

Conclusion: The data contain no direct information about the null-space component b_N. Conditional on b_R, the value of b_N is therefore entirely determined by the ℓ_1 penalty. Nevertheless, because the estimated b_R depends on the data, the lasso selected value of b_N can depend indirectly on the data through b_R.

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References

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